

Mohammad Ammar | AI/ML Research Engineer

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Portfolio LinkedIn

EDUCATION

Khoury College of Computer Sciences, Northeastern University , USA <i>M.S. in Artificial Intelligence</i> Related courses: Foundation of Artificial Intelligence, Algorithms, Reinforcement Learning	Sep 2025 – Dec 2027
Zakir Husain College of Engineering and Technology, AMU, India <i>B.Tech in Computer Engineering</i> Related courses: Data Structures, Algorithms, Distributed Computing, Computer Architecture, Applied Machine Learning	Jul 2020 – Apr 2024

TECHNICAL KNOWLEDGE

ML Frameworks:	Python, PyTorch, JAX, Ray RLLib, LangChain, Ollama, FastMCP
Model Optimization:	Pruning, Quantization, ONNX, DeepSparse, Hugging Face PEFT
Systems & Infrastructure:	CUDA, C++, ROS, Docker, Linux, Git, Ray, Gym, REST APIs
Data & Experimentation:	NumPy, SciPy, Pandas, TensorBoard, Hyperparameter Tuning, Weights & Biases

WORK EXPERIENCE

University of Kiel <i>Machine Learning Research Intern</i> - Developed MicroCrackAttentionNeXt, a deep learning model for micro-crack detection through acoustic signals, achieving 95.7% accuracy and 0.957 F1 on imbalanced data - Integrated attention and multi-scale representation learning with riemannian manifold-based visualizations to improve model performance and interpretability over CNN and transformer baselines	Jan 2025 - May 2025
University of North Dakota <i>Machine Learning Research Intern</i> - Built an RL-based UAV framework for GPS spoofing detection and mitigation, achieving 96%+ detection accuracy and 91%+ task success in adversarial conditions - Implemented a real-time DQN with a novel experience replay mechanism, delivering sub-25 ms inference latency and 6x faster learning	Jun 2024 – Jan 2025
Marine Technological Society - Autonomous Underwater Vehicle Club ZHCET <i>Computer Vision Engineer</i> - Spearheaded the development of custom datasets for vision tasks such as buoy detection and docking station recognition Gained 9x speedups over CPU baseline through utilizing PyTorch, ONNX and DeepSparse to prune, quantize and deploy the YOLOv5 model on Raspberry Pi 4	Jul 2022 – May 2023

PROJECTS

Custom Depth-Wise Separable Convolution Kernel - Designed and optimized a fused depthwise-separable convolution CUDA kernel integrated as a PyTorch C++ extension, eliminating intermediate global memory writes and reducing kernel launches - Achieved 2.76x inference speedup and 47% peak GPU memory reduction compared to cuDNN depthwise + pointwise baseline on NVIDIA GPU, with numerically exact outputs	Jan 2026 – Feb 2026
Fast-Tracking Drug Discovery with Generative AI - Fine-tuned the ESM-3 protein language model on a curated dataset of Designed Ankyrin Repeat Proteins (DARPin)s - Employed LoRA and DoRA PEFT techniques to achieve 98%+ accuracy while keeping the active training parameters under 4% of the total model size	Dec 2025 – Jan 2026
Multiagent Reinforcement Learning Aerial Dogfight - Engineered a high-throughput training pipeline leveraging Ray RLLib to orchestrate multi-agent PPO optimization across a distributed GPU cluster, training agents high level maneuvers from low level signals - Maximized hardware utilization by parallelizing data collection across multiple CPU cores to saturate GPU learners, completing 10,000+ episodes in an 18-hour window	Oct 2025 – Dec 2025

PEER-REVIEWED RESEARCH

- Reinforcement Learning Driven Integrated Detection and Mitigation of UAV GPS Spoofing Attacks – IEEE Internet of Things
- Optical Coherence Tomography Image Classification using Light-weight Hybrid Transformers – IEEE
- MicroCrackAttentionNeXt – MDPI Sensors
- From Rules to Pixels: A Decoupled Framework for Segmenting Human-Centric Rule Violations, MusIML Workshop NeurIPS